

YSP BUFENCON INJECTION

Ingredient(s):

Each ml contains:

Betamethasone Dipropionate	6.43mg
(eq. to Betamethasone 5.0mg)	
Betamethasone Sodium Phosphate	2.63mg
(eq. to Betamethasone 2.0mg)	
Benzyl Alcohol (as preservative)	5.0mg
Methyl Paraben (as preservative)	1.8mg
Propyl Paraben (as preservative)	0.2mg

Pharmacology (Summary of Pharmacodynamic and Pharmacokinetic):

Pharmacodynamic:

Betamethasone is a glucocorticoid that, when given systemically, has potent glucocorticoid activity and binds with high affinity to the glucocorticoid receptor. Betamethasone thus has catabolic effects on proteins in peripheral tissues, cause resistance to insulin and impair peripheral utilization of glucose, and immunosuppressive from bone. Betamethasone has both local anti-inflammatory and immunosuppressive activity. Thus, Betamethasone inhibits the adherence of neutrophils and monocyte macrophages to the capillary endothelial cells of the inflamed area. The drug blocks the effect of macrophage migration inhibitory factor and decreases the activation of plasminogen to plasmin. Finally, by inhibition of phospholipase A2 activity, via formation of lipocortin, Betamethasone reduces the formation of prostaglandins and leukotrienes in local tissue.

Pharmacokinetic:

1. Betamethasone decreases or prevents tissue responses to inflammatory processes, thereby reducing the development of symptoms of inflammation without affecting the underlying cause. It inhibits accumulation of inflammatory cells including macrophages and leukocytes at sites of inflammation.
2. Prompt therapeutic activity in corticosteroid-responsive conditions is achieved by the soluble ester, betamethasone sodium phosphate, which is quickly absorbed after injection. Sustained activity is provided by betamethasone dipropionate, which is only slightly soluble and affords a repository for slow absorption, thereby controlling symptoms over a prolonged period.
3. Betamethasone is primarily metabolized in the liver, other is in the kidney and tissue, to mostly inactive metabolites which are excreted in the urine.

Indication(s):

This product is indicated for the treatment of acute and chronic corticosteroid-responsive disorders. Corticosteroid hormone therapy is an adjunct to, and not a replacement for, conventional therapy.

Musculoskeletal and Soft Tissue Conditions: Rheumatoid arthritis; osteoarthritis; bursitis; ankylosing spondylitis; epicondylitis; radiculitis; coccydynia; sciatica; lumbago; torticollis; ganglion cyst; exostasis; fasciitis.

Allergic Conditions: Chronic bronchial asthma (including adjunctive therapy for status asthmaticus); hay fever; angioneurotic edema; allergic bronchitis; seasonal or perennial allergic rhinitis; drug reactions; serum sickness; insect bites.

Dermatologic Conditions: Atopic dermatitis (nummular eczema); neurodermatitis (circumscribed lichen simplex); contact dermatitis; severe solar dermatitis; urticaria; hypertrophic lichen planus; necrobiosis lipoidica diabetorum; alopecia areata; discoid lupus erythematosus; psoriasis; keloids; pemphigus; dermatitis herpetiformis; cystic acne.

Collagen Diseases: Disseminated lupus erythematosus; scleroderma; dermatomyositis; periarteritis nodosa.

Neoplastic Diseases: For palliative management of leukemias and lymphomas in adults; acute leukemia of childhood.

Other Conditions: Adrenogenital syndrome; ulcerative colitis; regional ileitis; sprue; podiatric conditions (bursitis under heloma durum, hallux rigidus, digiti quinti varus); affections requiring subconjunctival injection; corticosteroid-responsive blood dyscrasias; nephritis, and nephritic syndrome. Primary and secondary adrenocortical insufficiency may be treated with this product but should be supplemented with mineralocorticosteroids, if applicable.

This product is recommended for

- (1) intramuscular injection in conditions responsive to systemic corticosteroids;
- (2) injection directly into the affected soft tissues where indicated;
- (3) intraarticular and periarticular injection in arthritides;
- (4) intralesional injection in various dermatologic conditions; and
- (5) local injection in certain inflammatory and cystic disorders of the foot.

Dosage and Administration:

DOSING REQUIREMENTS ARE VARIABLE AND MUST BE INDIVIDUALIZED ON THE BASIS OF THE SPECIFIC DISEASE, ITS SEVERITY AND THE RESPONSE OF THE PATIENT.

The initial dose should be maintained or adjusted until a satisfactory response is observed. If a satisfactory clinical response does not occur after a reasonable period of time, treatment with this product should be discontinued and other appropriate therapy initiated.

Systemic Administration:

For systemic therapy, treatment is initiated with 1 to 2 ml in most conditions and repeated as necessary. Administration is by deep intramuscular (IM) injection in the gluteal region. Dosage and frequency of administration will depend on the severity of the patient's condition and the therapeutic response. 2 ml might be required initially in a severe illness, such as lupus erythematosus or status asthmaticus which has been resolved by appropriate life-saving procedures.

A wide variety of dermatologic conditions respond effectively to an IM injection of 1 ml injection, repeated according to the response of the conditions. In respiratory tract disorders, onset of relief from symptoms has occurred within a few hours after IM injection. Effective control of symptoms with 1 to 2 ml is obtained in bronchial asthma, hay fever, allergic bronchitis and allergic rhinitis. In the treatment of acute or chronic bursitis, excellent results are obtained with 1 to 2 ml IM injection of this product, repeated as necessary.

Local Administration:

Concomitant use of a local anesthetic is rarely necessary. If coadministration of a local anesthetic is desired, this product may be mixed (in a syringe, not the vial) with 1% or 2% procaine hydrochloride or lidocaine, using formulations which do not contain parabens. Similar local anesthetics may also be used. Anesthetics containing methylparaben, propylparaben, phenol, etc. should be avoided. The required dose of this product is first withdrawn from the vial into the syringe. The local anesthetic is then drawn in, and the syringe is shaken briefly.

- In acute subdeltoid, subacromial, olecranon, and prepatellar bursitis, an intrabursal injection of 1 to 2ml of this product may relieve pain and restore full range of movement within a few hours. Chronic bursitis may be treated with reduced dosage once acute symptoms are controlled.
- In acute tenosynovitis, tendinitis, and peritendinitis, one injection of this product should alleviate the condition. In chronic forms of these conditions, it may be necessary to repeat the injection as the patient's condition requires.
- Following 0.5 to 2 ml intra-articular administration of this product, relief from pain, soreness, and stiffness associated with rheumatoid arthritis and osteoarthritis may be experienced within two to four hours. Duration of relief, which varies widely in both diseases, is four or more weeks in the majority of cases. An intra-articular injection of this product is well tolerated in the joint and periarticular tissues. Recommended doses for intra-articular injection are: large joints (knee, hip, shoulder), 1 to 2ml; medium joints (elbow, wrist, ankle), 0.5 to 1 ml; small joints (foot, hand, chest), 0.25 to 0.5 ml.
- Dermatologic conditions may respond to intralesional administration of this product. Response of some lesions not treated directly may be due to a slight systemic effect of the drug. In intralesional treatment, an intradermal dosage of 0.2 ml/cm² of this product evenly injected with a tuberculin syringe and a 26-gauge needle is recommended. The total amount of this product injected at all sites each week should not exceed 1 ml.
- This product may be used effectively in disorders of the foot that are responsive to corticosteroid therapy. Bursitis under heloma durum may be controlled with two successive injections of 0.25 ml each. In some conditions, such as hallux rigidus, digiti quinti varus and acute gouty arthritis, onset of relief may be rapid. A tuberculin syringe with a 25-gauge needle is suitable for most injections. Recommended doses at intervals of approximately one week: bursitis under heloma durum or molle, 0.25 to 0.5 ml; bursitis under calcaneal spur, 0.5 ml; bursitis over hallux rigidus, 0.5 ml; bursitis over digiti quinti varus, 0.5 ml; synovial cyst, 0.25 to 0.5 ml; Morton's neuralgia (metatarsalgia), 0.25 to 0.5 ml; tenosynovitis, 0.5 ml; periostitis of cuboid, 0.5 ml; acute gouty arthritis, 0.5 to 1 ml.

After a favorable response is obtained, the proper maintenance dosage should be determined by decreasing the initial dose in small decrements at appropriate time intervals until the lowest dose which will maintain an adequate clinical response is determined.

Exposure of the patient to stressful situations unrelated to the existing disease may necessitate an increased dose of this product. If the drug is to be discontinued after long-term therapy, the dose should be decreased gradually.

Mode of Administration:

I.M. and Local Administration(Intra-articular Injection, Intradermal Injection, Intralesional Injection)

Contraindications:

Contraindicated in patients hypersensitive to Betamethasone or other Corticosteroid and any components of this product. As with other corticosteroids should not be injected into unstable joints, infected areas, or intervertebral space.

Precaution(s) / Warning(s):

1. This product is not for intravenous or subcutaneous use.
2. Strict aseptic technique is mandatory in the use of this product.
3. This product contains two betamethasone esters, one of which, betametasone disodium phosphate, disappears rapidly from the injection site. The potential for systemic effects produced by this soluble portion of this product should therefore be taken into account by the physician when using this preparation.
4. This product should be administered intramuscularly with caution to patients with idiopathic thrombocytopenic purpura.
5. IM injections of corticosteroids should be given deep into the large muscle masses to avoid local tissue atrophy. Soft tissue, intralesional and intra-articular administration of a corticosteroid may produce systemic as well as local effects.
6. Examination of any joint fluid present is necessary to exclude a septic process.
7. Local injection into a previously infected joint is to be avoided. A marked increase in pain and local swelling, further restriction of joint motion, fever and malaise are

suggestive of septic arthritis. If the diagnosis of sepsis is confirmed, appropriate antimicrobial therapy should be instituted.

8. Corticosteroids should not be injected into unstable joints, infected areas or intervertebral spaces. Repeated injections into joints of osteoarthritis may increase joint destruction. Avoid injecting corticosteroids directly into the substance of tendons because delayed appearance of tendon rupture has resulted.
9. Following intra-articular corticosteroid therapy, care should be taken by the patient to avoid overuse of the joint in which symptomatic benefit has been obtained. Because rare instances of anaphylactoid reactions have occurred in patients receiving parenteral corticosteroid therapy, appropriate precautionary measure should be taken prior to administration, especially when the patient has a history of allergy to any drug. With long-term corticosteroid therapy, transfer from parenteral to oral administration should be considered after weighing the potential benefits and risks.
10. Dosage adjustments may be required with remission or exacerbation of the disease process, the patient's individual response to therapy and exposure of the patient to emotional or physical stress such as serious infection, surgery or injury.
11. Monitoring may be necessary for up to one year following cessation of long-term or high-dose corticosteroid therapy. Corticosteroids may mask some signs of infection, and new infections may appear during use. When corticosteroids are used, decreased resistance and inability to localize infection may occur.
12. Prolonged corticosteroid use may produce posterior subcapsular cataracts (especially in children), glaucoma with possible damage to the optic nerves, and may enhance secondary ocular infections due to fungi or viruses.
13. Average and large doses of corticosteroids can cause elevation of blood pressure, salt and water retention, and increased excretion of potassium. These effects are less likely to occur with the synthetic derivatives except when used in large doses. Dietary salt restriction and potassium supplementation may be considered.
14. All corticosteroids increase calcium excretion.
15. While on corticosteroid therapy patients should not be vaccinated against smallpox. Other immunization procedures should not be undertaken in patients receiving corticosteroids, especially high doses, because of possible hazards of neurological complications and lack of antibody response. However, immunization procedures may be undertaken in patients who are receiving corticosteroids as replacement therapy, e.g., for Addison disease.
16. Patients who are on immunosuppressant doses of corticosteroids should be warned to avoid exposure to chickenpox or measles and, if exposed, to obtain medical advice. This is of particular importance in children.
17. Corticosteroid therapy in active tuberculosis should be restricted to those cases of fulminating or disseminated tuberculosis in which the corticosteroid is used for management in conjunction with an appropriate antituberculous regimen.
18. If corticosteroids are indicated in patients with latent tuberculosis or tuberculin reactivity, close observation is necessary since reactivation of the disease may occur.
19. During prolonged corticosteroid therapy, patients should receive chemoprophylaxis. If rifampin is used in a chemoprophylactic program, its enhancing effect on metabolic hepatic clearance of corticosteroids should be considered; adjustment in corticosteroid dosage may be required.
20. The lowest possible dose of corticosteroid should be used to control the condition under treatment; when dosage reduction is possible, it should be gradual. Drug-induced secondary adrenocortical insufficiency may result from too rapid corticosteroid withdrawal and may be minimized by gradual dosage reduction.
21. Such relative insufficiency may persist for months after discontinuation of therapy; therefore, if stress occurs during that period, corticotherapy should be reinstituted. If the patient is receiving corticosteroid already, dosage may have to be increased.
22. Since mineralocorticoid secretion may be impaired, salt and/or a mineralocorticosteroid should be administered concurrently. Corticosteroid effect is enhanced in patients with hypothyroidism or in those with cirrhosis.
23. Cautious use of corticosteroids is advised in patients with ocular herpes simplex because of possible corneal perforation. Psychic derangements may appear with corticosteroid therapy. Existing emotional instability or psychotic tendencies may be aggravated by corticosteroids.
24. Corticosteroids should be used with caution in: nonspecific ulcerative colitis, if there is a probability of impending perforation, abscess, or other pyogenic infection; diverticulitis; fresh intestinal anastomoses; active or latent peptic ulcer; renal insufficiency; hypertension; osteoporosis; and myasthenia gravis.
25. Since complications of glucocorticosteroid treatment are dependent on dose, size and duration of treatment, a risk/benefit decision must be made with each patient.
26. Since corticosteroid administration can disturb growth rates and inhibit endogenous corticosteroid production in infants and children, the growth and development of these patients receiving prolonged therapy should be followed carefully.
27. Corticosteroids may alter the motility and number of spermatozoa in some patients.
28. Pheochromocytoma crisis, which may be fatal, has been reported after administration of systemic corticosteroids. Corticosteroids should only be administered to patients with suspected or identified pheochromocytoma after an appropriate risk/benefit evaluation.

As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age. Not to be used in neonates.

Interaction with Other Medicaments:

1. Concurrent use of phenobarbital, phenytoin, rifampin or ephedrine may enhance the metabolism of corticosteroids, reducing their therapeutic effects.
2. Patients receiving both a corticosteroid and an estrogen should be observed for excessive corticosteroid effects.
3. Concurrent use of corticosteroids with potassium-depleting diuretics may enhance

hypokalemia. Concurrent use of corticosteroids with cardiac glycosides may enhance the possibility of arrhythmias or digitalis toxicity associated with hypokalemia. Corticosteroids may enhance the potassium depletion caused by amphotericin B. In all patients taking any of these drug therapy combinations, serum electrolyte determinations, particularly potassium levels, should be monitored closely.

4. Concurrent use of corticosteroids with coumarin-type anticoagulants may increase or decrease the anticoagulant effects, possibly requiring adjustment in dosage.
5. Combined effects of non-steroidal anti-inflammatory drugs or alcohol with glucocorticosteroids may result in an increased occurrence or increased severity of gastrointestinal ulceration.
6. Corticosteroids may decrease blood salicylate concentrations. Acetylsalicylic acid should be used cautiously in conjunction with corticosteroids in hypothermia.
7. Dosage adjustment of an antidiabetic drug may be necessary when corticosteroids are given to diabetics.
8. Concomitant glucocorticosteroid therapy may inhibit the response to somatotropin.
9. Co-treatment with CYP3A inhibitors, including cobicistat-containing products, is expected to increase the risk of systemic side-effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects, in which case patients should be monitored for systemic corticosteroid side-effects.

Pregnancy and Lactation:

Since adequate human reproduction studies have not been studied with corticosteroids, the use of this drug in pregnancy, nursing mothers, or women of child bearing potential requires that the risk benefit ratio should be considered.

Side Effect(s) / Adverse Reaction(s):

Fluid and electrolyte disturbances: sodium retention; congestive heart failure in susceptible patients; potassium loss; hypokalemic alkalosis; hypertension.

Musculoskeletal: muscle weakness; steroid myopathy; loss of muscle mass; osteoporosis; vertebral compression fractures; aseptic necrosis of femoral and humeral heads; pathologic fractures of long bones.

Gastrointestinal: peptic ulcer with possible subsequent perforation and hemorrhage; pancreatitis; abdominal distention; ulcerative esophagitis.

Dermatological: impaired wound healing; thin fragile skin; petechiae and ecchymoses; facial erythema; increased sweating; may suppress reaction to skin test.

Neurological: convulsions; increased intracranial pressure with papilledema (pseudotumor cerebri) usually after treatment; vertigo; headache.

Endocrine: menstrual irregularities; development of cushingoid state; suppression of growth in children; secondary adrenocortical and pituitary unresponsiveness, particularly in times of stress, trauma, surgery, or illness; decreased carbohydrate tolerance; manifestation of latent diabetes mellitus; increased requirements for insulin or oral hypoglycemic agents in diabetics.

Ophthalmologic: posterior sub-capsular cataracts; increased intraocular pressure; glaucoma; exophthalmos.

Metabolic: negative nitrogen balance due to protein catabolism.

The following adverse reactions are related to parenteral corticosteroid therapy. Rare instances of blindness associated with intralesional therapy around the face and the head; Hyper-pigmentation or hypo-pigmentation; subcutaneous and cutaneous atrophy; sterile abscess; post-injection flare (following intra-articular use); Charcot-like arthropathy.

Symptoms and Treatment for Overdosage, and Antidote(s):

A single overdose of a corticosteroid would not be expected to produce acute symptoms. Hypercorticotoid effects are not anticipated unless there has been repeated administration of high doses.

Systemic effects of chronic overdosage with steroids include effects on sodium and water retention, increased appetite, mobilization of calcium and phosphorus with osteoporosis, nitrogen depletion, hyperglycemia, effects on tissue repair, increased susceptibility to infection, adrenal insufficiency, adrenal cortex hyperactivity, mental and neurological disturbances and muscular weakness.

In case of an acute overdose, maintain adequate fluid intake and monitor electrolytes in serum and urine, with particular attention to sodium and potassium balance. In case of chronic toxicity, slowly withdraw the drug. Treat electrolyte imbalance if necessary.

Shelf-Life:

3 years from the date of manufacture.

Storage Condition(s):

Store at temperature below 30°C.

Protect from light and moisture.

Incompatibilities:

Do not mix with diluents or local anesthetics containing preservatives (eg. paraben, phenol) because flocculation of suspension may result

Product Description:

A white to slightly yellowish white suspension.

Dosage Forms and Packaging Available:

5ml x 10, 5ml x 50 and 5ml x 100 vials.



Manufacturer and Product Registration Holder :
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